Exam Project – Copenhagen on Two Wheels

Mathematical Modelling
02526

The goal of this project is to use mathematical modelling to solve a practical problem: Predicting bicycle traffic volumes in Copenhagen based on weather, calendar, and temporal features. You should:

- Identify significant issues and patterns in the data
- Formulate mathematical models that address these issues
- Implement the models in a computer program
- Use the implemented models for analysis and prediction
- Report the results from the analysis and discuss the validity of the models

Figure 1: **Bicycle rush hour in Copenhagen.** On a typical weekday, thousands of cyclists pass through the city's busiest corridors every hour. *How many will ride tomorrow?* [The image was generated with Nano Banana Pro]

The report for this project must be **5 pages at most**, including graphs, tables, and images (excluding the front page and appendices). In this exercise, several questions are asked. The report, however, should not be a list of answers, but instead be a coherent documentation and discussion of the analysis performed.

The project consists of several parts. First, you will explore the data and understand the patterns in bicycle traffic. Then, you will build regression models of increasing sophistication to predict traffic volumes. Finally, you will evaluate your models and discuss their strengths and limitations.

To solve the project, we suggest you use PYTHON.

Background

Copenhagen is one of the world's leading cycling cities. Over 380 km of dedicated cycle tracks serve the city, and more than 40% of commuters travel by bicycle. The municipality operates a network of automated bicycle counting stations at key locations, such as bridges, major streets, and cycling corridors. They record the number of passing bicycles every hour throughout the day. Every day, 42,600 cyclists cross Dronning Louises Bro alone, which is more than three times as many as cars crossing the Øresund Bridge between Denmark and Sweden.

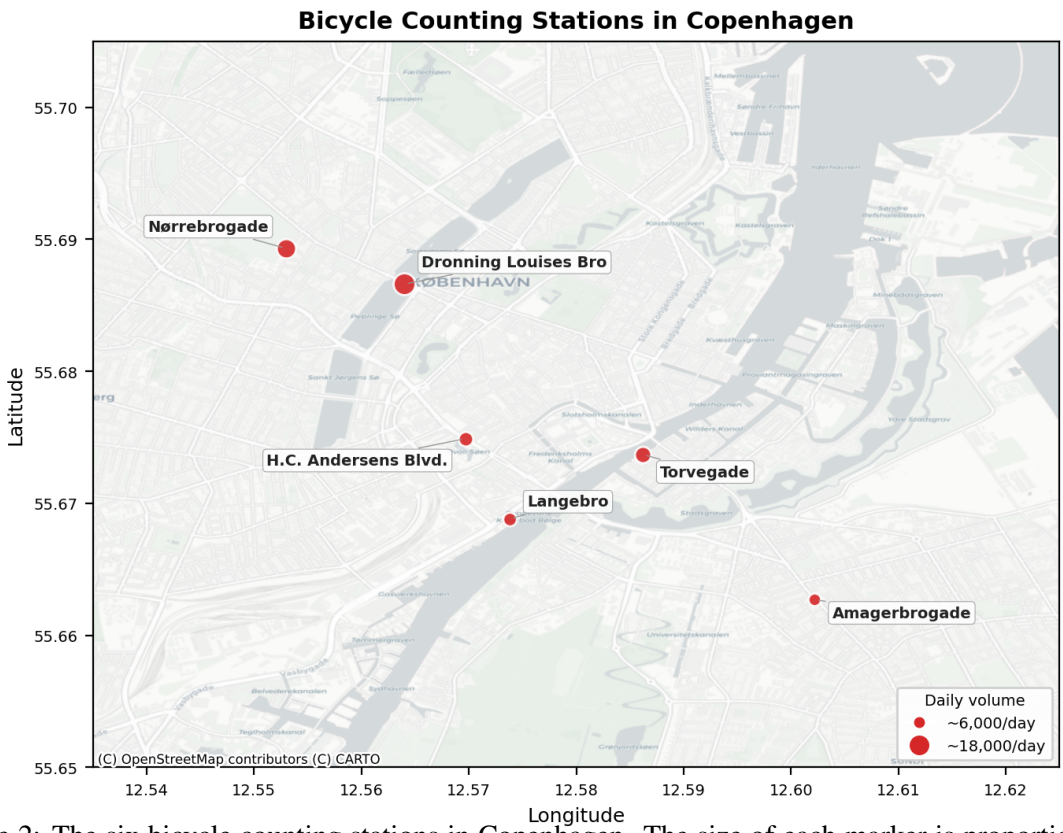

Figure 2: The six bicycle counting stations in Copenhagen. The size of each marker is proportional to the average daily traffic volume. Dronning Louises Bro is the busiest cycling location in Denmark.

Understanding what drives cycling volumes is important for traffic planning, infrastructure investment, real-time traffic management, and environmental impact evaluation. In particular, the ability to *predict* bicycle traffic in the near future, given a weather forecast and the calendar, would help city planners allocate resources and manage the cycling infrastructure more effectively.

In this project, you will work with hourly bicycle count data from 6 counting stations in Copenhagen together with weather observations from the same period. Your task is to build mathematical models that can predict bicycle traffic volume from weather and calendar features. The six counting stations are shown in Fig. 2.

Data

You are given hourly data for 3 years (2021–2023) from 6 counting stations, together with hourly weather observations from the Copenhagen area. The data files are:

- `bike_counts.csv` — Hourly bicycle counts per station (long format). Approximately 2% of values are missing due to sensor outages.
- `weather_copenhagen.csv` — Hourly temperature (°C), wind speed (m/s), and precipitation (mm).
- `stations.csv` — Metadata for the 6 stations (name, coordinates, typical daily volume).
- `danish_holidays.csv` — Dates of Danish public holidays in the period.

Fig. 3 and 4 show examples of the data patterns you will encounter.

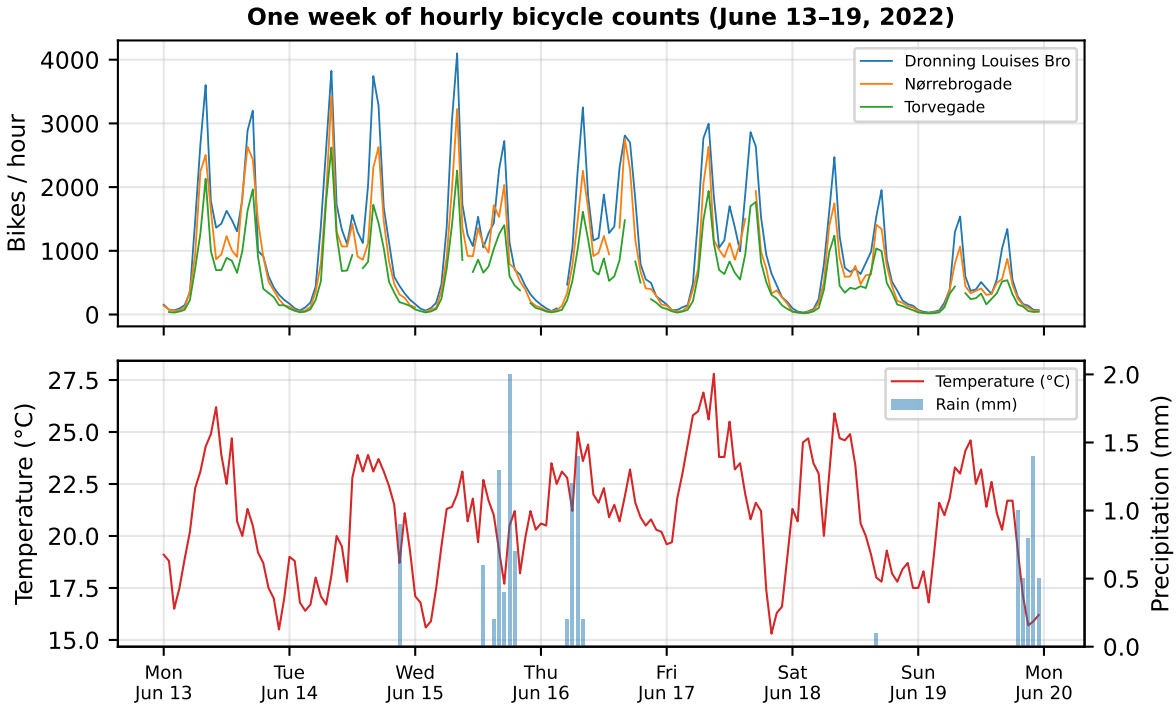

Figure 3: **One week of hourly data (June 13–19, 2022).** Top: bicycle counts for three stations showing clear commuting peaks on weekdays and lower, smoother weekend patterns. Bottom: temperature and precipitation during the same period.

Mathematical Models

Linear Regression

We want to predict the bicycle count y at a given station and hour from a feature vector $\mathbf{x} \in \mathbb{R}^d$ containing weather and calendar information. A natural starting point is the linear model:

$$y = \mathbf{x}^T \boldsymbol{\beta} + \varepsilon$$

where $\boldsymbol{\beta} \in \mathbb{R}^d$ is the parameter vector and ε represents noise. We assume that

$$\begin{aligned} \mathrm{E}(\boldsymbol{\varepsilon}) &= \mathbf{0} \\ (\boldsymbol{\varepsilon}) &= \sigma^2 \mathbf{I}_n. \end{aligned}$$

Given N observations stacked into a matrix $\mathbf{X} \in \mathbb{R}^{N \times d}$ and a vector $\mathbf{y} \in \mathbb{R}^N$, the ordinary least squares (OLS) estimator assumes that $\boldsymbol{x}$ has maximum rank:

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

This minimizes the sum of squared residuals: $\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2$.

Furthermore, we assume that the independent variables are scaled so that $\mathbf{X}^T\mathbf{X}$ has correlation form (i.e. the single independent variables are reduced by their average and divided by their standard deviation). This normalization will help make the estimates more stable numerically and is often recommendable in a practical situation.

If $\mathbf{X}^T\mathbf{X}$ in this form is close to a unity matrix, i.e. if the independent variables are near-orthogonal, the least squares estimator is fine. If we have ***multicollinearity***, i.e. if the independent variables are strongly correlated, the estimates $\hat{\boldsymbol{\beta}}$ will be unstable.

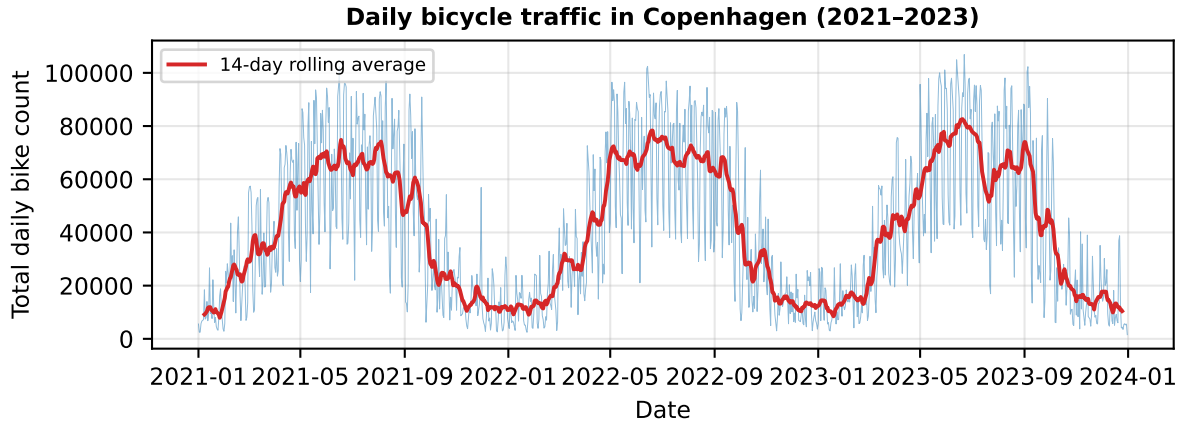

Figure 4: **Total daily bicycle count across all stations for the full 3-year period.** Summer volumes are roughly 2-3 times larger than winter. The red line shows a 14-day rolling average.

The role of eigenvalues

We now analyze some properties of $\hat{\boldsymbol{\beta}}$ to understand when and why OLS becomes unstable. We have

$$\mathrm{V}(\hat{\boldsymbol{\beta}}) = \sigma^2(\mathbf{X}^T\mathbf{X})^{-1}.$$

If we put L equal to the distance from $\hat{\boldsymbol{\beta}}$ to $\boldsymbol{\beta}$,

$$L^2 = (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})^T(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}),$$

we get

$$\mathrm{E}(L^2) = \sum_{i=1}^{p} \mathrm{E}[(\hat{\beta}_i - \beta_i)^2] = \sum_{i=1}^{p} \mathrm{V}(\hat{\beta}_i) = \sigma^2 \operatorname{tr}(\mathbf{X}^T\mathbf{X})^{-1}.$$

Since $L^2 = \hat{\boldsymbol{\beta}}^T\hat{\boldsymbol{\beta}} - 2\hat{\boldsymbol{\beta}}^T\boldsymbol{\beta} + \boldsymbol{\beta}^T\boldsymbol{\beta}$, we get that the expected value of the squared length of $\hat{\boldsymbol{\beta}}$ is

$$\mathrm{E}(\hat{\boldsymbol{\beta}}^T\hat{\boldsymbol{\beta}}) = \sigma^2 \operatorname{tr}(\mathbf{X}^T\mathbf{X})^{-1} + \boldsymbol{\beta}^T\boldsymbol{\beta}.$$

If we denote the eigenvalues of $\mathbf{X}^T\mathbf{X}$ by $\lambda_1 \geq \cdots \geq \lambda_p$, we obtain

$$\mathrm{E}(L^2) = \sigma^2\left(\frac{1}{\lambda_1} + \cdots + \frac{1}{\lambda_p}\right) > \frac{\sigma^2}{\lambda_p},$$

and

$$\mathrm{E}(\hat{\boldsymbol{\beta}}^T\hat{\boldsymbol{\beta}}) = \sigma^2\left(\frac{1}{\lambda_1} + \cdots + \frac{1}{\lambda_p}\right) + \boldsymbol{\beta}^T\boldsymbol{\beta} > \frac{\sigma^2}{\lambda_p} + \boldsymbol{\beta}^T\boldsymbol{\beta}.$$

If the independent variables are strongly correlated, the eigenvalues of $\mathbf{X}^T\mathbf{X}$ will vary a lot and consequently the smallest will be very small ($\ll 1$). According to the relations above, the squared distance between $\boldsymbol{\beta}$ and $\hat{\boldsymbol{\beta}}$ will in this case have a large expected value, and the squared length of $\hat{\boldsymbol{\beta}}$ will have an expectation by far exceeding the squared length of $\boldsymbol{\beta}$.

This tendency to inflation of $\hat{\boldsymbol{\beta}}$ is caused by requiring unbiasedness. The question is therefore whether we, by relaxing on this requirement, may obtain estimates that in some sense are closer to $\boldsymbol{\beta}$.

The bias-variance tradeoff and ridge regression

Here we may refer to the ***mean squared error*** of an estimator $\tilde{\beta}$ (in the one-dimensional case):

$$\mathrm{MSE}(\tilde{\beta}) = \mathrm{E}[(\tilde{\beta} - \beta)^2] = \mathrm{V}(\tilde{\beta}) + \{\mathrm{E}(\tilde{\beta}) - \beta\}^2,$$

i.e. that the MSE of an estimator is equal to the variance plus the squared bias. If we therefore by allowing a small bias may obtain a great reduction in the variance, this would obviously be preferable. This is exactly what is obtained with the ***ridge estimator***.

A ridge estimator for $\boldsymbol{\beta}$ in the model $\boldsymbol{Y} = \mathbf{X}\,\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ is an estimator $\hat{\boldsymbol{\beta}}_k^* = \hat{\boldsymbol{\beta}}^*$, that is a solution to

$$(\mathbf{X}^T\mathbf{X} + k \cdot \mathbf{I})\hat{\boldsymbol{\beta}}^* = \mathbf{X}^T\boldsymbol{Y},$$

i.e.

$$\hat{\boldsymbol{\beta}}^* = (\mathbf{X}^T\mathbf{X} + k \cdot \mathbf{I})^{-1}\mathbf{X}^T\boldsymbol{Y}.$$

Here $k \geq 0$ is the regularization parameter. By adding $k \cdot \mathbf{I}$ to $\mathbf{X}^T\mathbf{X}$, the smallest eigenvalues are shifted away from zero, which stabilizes the inversion. The price is a small bias in the estimates. In numerical mathematics, this is called a ***(Tikhonov) regularization*** and is a very common way of solving ill-posed problems.

You should test multiple values of k and select the best using the validation set.

Note that there are other alternatives to regularisation than the Ridge. Whatever you use, you should understand it!

Feature Engineering

The raw features (temperature, wind speed, precipitation, hour, day of week, month) may not capture the patterns well in a linear model. You will likely need to engineer better features. Consider for example:

- Encoding cyclical features (hour, day of week, month) using sine/cosine transforms.
- Polynomial or interaction terms (e.g. temperature $\times$ hour).
- Indicator variables for weekday/weekend and holidays.
- Non-linear transformations of weather variables.

Project Tasks

Data split. Split your data into train/validation/test sets. For example, you can split them chronologically and train on 01-01-2021 to 30-06-2022, validate on 01-07-2022 to 31-12-2022, and hold out 2023 as the final test set.

Part 1 — Exploratory Data Analysis

This part is for your own understanding and should inform the report.

- Load the data and plot the various activity curves for one or more stations. Get a feeling for the patterns.
- Investigate the hourly, weekly, and seasonal patterns. Are they consistent across stations?
- Investigate the relationship between weather variables and bicycle counts. Which variables appear most influential?
- How should missing values be handled? Document your choice.

Part 2 — Baseline Model

- Choose one station to focus on initially (e.g. Dronning Louises Bro).
- Build a simple linear regression model using raw features: temperature, wind speed, precipitation, hour, day of week, and month.
- Evaluate the model on the validation set using appropriate metrics (e.g. RMSE, R^2, MAE). Visualize predictions vs. actual counts.
- Examine the predicted errors. Are there systematic patterns that suggest the model is missing something?

Part 3 — Improved Model

- Improve your feature representation. Consider some of the transformations suggested in the section on feature engineering and your own ideas.
- Apply ridge regression (or another regularization method) and select the regularization parameter using the validation set.
- Compare the improved model to the baseline. Quantify the improvement and discuss which features matter most.
- Visualize the learned weights/coefficients and interpret them. Do they make sense?

Part 4 — Generalization and Analysis

- Apply your best model to the held-out test set (year 2023). Report the performance and compare it to the validation performance.
- Train and evaluate the model for *all six stations*. Can one model work for all stations, or do different stations need different models? Visualize and discuss.
- Investigate if a model trained on one station transfers to another (i.e. train on Dronning Louises Bro, predict Torvegade). What does this tell you about the stations?

Part 5 — Choose one extension

Choose **one** of the following extensions. It is fine to pick just one — going deeper on one topic is better than scratching the surface of both. It might be difficult to have a sufficient description of both within the page limit.

5.A — Classification: Can you classify whether a given hour will be "high traffic" or "low traffic" without knowing the actual count? Define a sensible threshold (e.g. median hourly count), build a linear classifier (e.g. logistic regression or LDA), and evaluate its accuracy, precision, and recall. Compare the classifier's decision boundary to the regression model's predictions.

5.B — Temporal prediction: Instead of predicting traffic from weather alone, investigate whether including *past traffic counts* (e.g. the count 1 hour ago, 24 hours ago, or 1 week ago) improves prediction. This turns the problem into a time series forecasting task. Discuss the practical implications. For example, the weather model only needs a weather forecast, while the temporal model needs recent real-time data.

Reporting Contents of the report

1. Describe the problem and the background – what is modelled and why?

2. Describe data and experiments
3. Describe the mathematical model - how and why?
4. Describe results
5. Discuss results – how good are the results? To what degree do they reflect the true problem?
6. Conclude – what is the contribution of the analysis?

Hand in The report is uploaded to DTU Learn. The code used (`*.py` files) must also be uploaded. The code must also be included in an appendix in the report.

References

1. Copenhagen Municipality, *The Bicycle Account 2022 — Copenhagen City of Cyclists*, 2022. https://kk.sites.itera.dk/apps/kk_pub2/pdf/2420_d4db2492337f.pdf
2. Copenhagen Municipality Open Data, Fixed bicycle counts (Faste cykeltællinger): https://www.opendata.dk/city-of-copenhagen/faste-cykeltaellinger
3. Danish Meteorological Institute, Open Data API: https://opendatadocs.dmi.govcloud.dk/
4. K. Conradsen, A.N. Christensen, A.A. Nielsen, B.K. Ersbøll, *Multivariate Statistics for the Technical Sciences*, V0.94, Kgs. Lyngby, 2018.

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